BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 077 · Testing: Normal, Boundary and Invalid Data

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Why use meaningful variable names?

Q2. What is FTP used for?
A. Transferring files
B. Sending email
C. Looking up domain names
D. Assigning IP addresses

Q3. “Malware means any program that uses a large amount of RAM.” Correct this. Explain why.

Q4. A school website is flooded with requests. “A denial-of-service attack makes access better for real users.” Correct this. What happens to them?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Normal test data — Representative valid inputs normally expected during use.

Boundary test data — Values at, immediately below and immediately above an allowed limit.

Invalid or erroneous — Inputs rejected by the required rules, including out-of-range or wrong-format data.

Iterative and final testing — Iterative testing runs during development; terminal testing checks the finished solution.

Test case — An input and expected outcome used to check correctness.

Bug — A defect in the program.

New learning

Normal test data: Representative valid inputs normally expected during use. Consider this in a test a marks form.

Boundary test data: Values at, immediately below and immediately above an allowed limit. Consider this in a ticket quantity checker.

Invalid or erroneous: Inputs rejected by the required rules, including out-of-range or wrong-format data. Consider this in a age-restricted service.

Iterative and final testing: Iterative testing runs during development; terminal testing checks the finished solution. Consider this in a sports score input.

Worked example

A test a marks form needs normal test data and boundary test data. Explain why. [4 marks]

1. Identify the inputs, constraints and expected result.

2. Normal test data [1]; Representative valid inputs normally expected during use. [1].

3. Boundary test data [1]; Values at, immediately below and immediately above an allowed limit. [1].

4. Give a specific point and a linked reason for each feature.

We Do · Practise with guidance

Guided 1

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

In a ticket quantity checker, explain the purpose of normal test data. [2 marks]

Worked method: Normal test data [1]; Representative valid inputs normally expected during use. [1].

Guided 2

A age-restricted service needs boundary test data. Explain its role. [2 marks]

Worked method: Boundary test data [1]; Values at, immediately below and immediately above an allowed limit. [1].

Guided 3

A sports score input uses invalid or erroneous and iterative and final testing. Explain both. [4 marks]

Worked method: Invalid or erroneous [1]; Inputs rejected by the required rules, including out-of-range or wrong-format data. [1]. Iterative and final testing [1]; Iterative testing runs during development; terminal testing checks the finished solution. [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Change the test value to the highest valid unsigned 8-bit integer.

```
value = 256
if value >= 0 AND value <= 255 then
    print("Accepted")
else
    print("Rejected")
endif
```

test

Question 1

State TWO concepts relevant to this topic. [2 marks] Source: Original OCR J277-style teaching question; not an official past paper.

Question 2

Explain how boundary test data and invalid or erroneous help in a club attendance tool. [4 marks] Source: Original OCR J277-style teaching question; not an official past paper.

Question 3

Explain the roles of normal test data, invalid or erroneous and iterative and final testing in a online purchase form. [6 marks] Source: Original OCR J277-style teaching question; not an official past paper.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Exit Ticket · Answer before checking

Normal test data contains valid inputs that are typically expected during use.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Clear names show what values mean and help people read and change code.

Q2. A — Transferring files

Q3. Malware is made to cause harm, disruption or access without permission. Different types work in different ways.

Q4. A denial-of-service attack floods a service with traffic or requests. Genuine users then cannot use it.

I Do / We Do pages · Guided practice

Q1. Normal test data [1]; Representative valid inputs normally expected during use. [1].

Q2. Boundary test data [1]; Values at, immediately below and immediately above an allowed limit. [1].

Q3. Invalid or erroneous [1]; Inputs rejected by the required rules, including out-of-range or wrong-format data. [1]. Iterative and final testing [1]; Iterative testing runs during development; terminal testing checks the finished solution. [1].

You Do pages · Code First

```
value = 255
if value >= 0 AND value <= 255 then
    print("Accepted")
else
    print("Rejected")
endif
```

Expected result: Accepted

You Do pages · Independent questions

Q1. Normal test data [1]; Invalid or erroneous [1].

Q2. Boundary test data [1]; Values at, immediately below and immediately above an allowed limit. [1]. Invalid or erroneous [1]; Inputs rejected by the required rules, including out-of-range or wrong-format data. [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Q3. Normal test data [1]; Representative valid inputs normally expected during use. [1]. Invalid or erroneous [1]; Inputs rejected by the required rules, including out-of-range or wrong-format data. [1]. Iterative and final testing [1]; Iterative testing runs during development; terminal testing checks the finished solution. [1].

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: